

b.Tech Mini-project seminar on
AUTORECORDER: INTELLIGENT EVENT DATA RECORDER FOR
VEHICLES

By
K.Sai Krishna(22831A0452)

Under the Supervision of
Mr. M. RAMESH NAIK
(Assistant Professor)



Department of electronics and communication
engineering,
Guru Nanak institute of technology

contents

- Abstract
- Introduction
- Literature Survey and Research gap
- Problem Identified
- Methodology
- Advantages and Disadvantages
- Applications
- Result
- Conclusion
- References

abstract

- ▶ The main purpose of this wireless black box project is to develop a vehicle black box system that can be installed into any vehicle all over the world.
- ▶ This paradigm is often designed with minimum range of circuits. Wireless black box is basically a device that will indicate all the parameters of a vehicle crash and will also store and display its parameters at every three seconds such as date, time, temperature, location, vibration, alcohol limit etc.
- ▶ At the time of accident or abnormal states of sensors, the message will be sent from the system built inside the car to the registered mobile numbers such as emergency numbers of police stations, hospitals, family members, owner etc.
- ▶ We have used various types of sensors like temperature sensor, Vibration sensor measures vibrations felt by the car during accident.

Introduction

Introduction: An Intelligent Event Data Recorder (EDR) is like a black box for vehicles. It records important data such as speed, braking, location, and impact during accidents.

- ▶ Unlike basic EDRs, the intelligent version uses sensors and GPS to collect and send data in real time.
- ▶ This helps in understanding accidents, improving safety, and making driving more responsible.

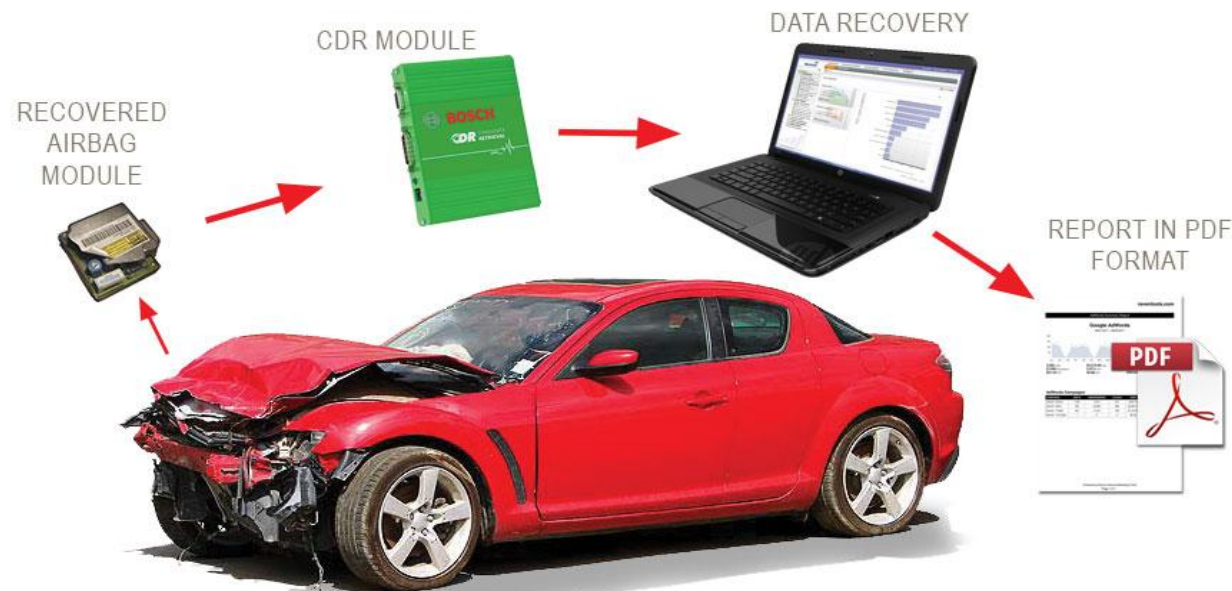


Fig 1. event data recorder in vehicle

Literature survey

1. Title: Driver Distraction Detection Using Event Cameras.

- Authors: Waseem Shariff and team.
- Year: 2024.
- Focus: They used a special camera and brain like AI to spot when drivers are distracted.
- Research Gap: It works well but it needs more testing in real life driving and different weather or lighting.

2. Title :Testing EDRs Without Crashing a Vehicle.

Authors: Matthew DiSogra and team.

- Year: 2024.
- Focus: They made away to test EDRs by simulating vehicle speed instead of crashig cars.
- Research Gap: Only simulates speed doesn't include all the sensors a real crash would use.

3. Title: Seatbelt Detection System.

- Authors: Paul Kielty and team.
- Year: 2023.
- Focus: They created a system that checks if you are wearig a seatbelt using a small smart camera.
- Research Gap: Might not work the same in all the cars or lighting conditions.

4. Title: Better Tracking With Event Cameras.

- Authors:Sumin Hu and team.
- Year: 2022.
- Focus: They made asystem that tracks moving things quickly, use ful for cars that record events.
- Research Gap: Needs to be tested in busy or complex driving situation.

Problem identified

- Still need more real-world testing.
- Not properly integrated with other technologies.
- Should be made simpler to use in all vehicles.
- Needs high processing power.
- Focused on only one function.
- Develop better testing methods.

methodology

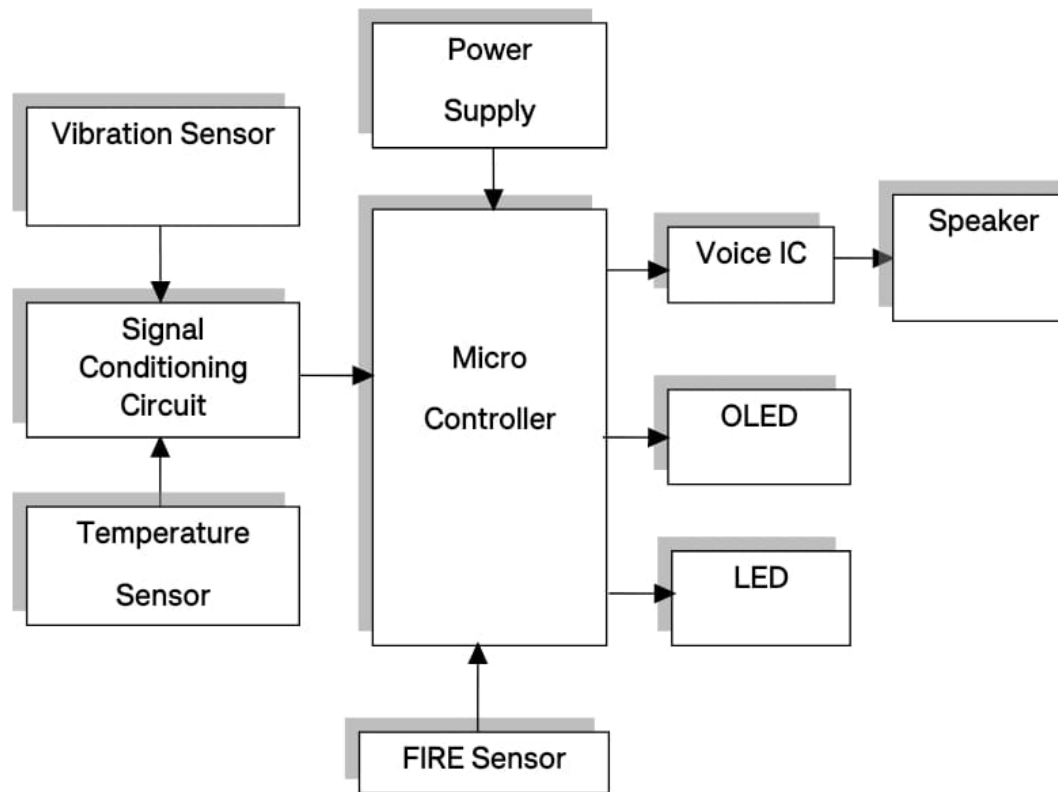


Fig 2. Block Diagram of Proposed Model

Advantages

- ▶ Helps in accident cases
- ▶ Makes driving safer
- ▶ Checks drivers actions
- ▶ Useful for insurance and police
- ▶ Detects vehicles problems early
- ▶ Encourages better driving



disadvantages

- ▶ Privacy issues
- ▶ It can be expensive
- ▶ Risk of hacking
- ▶ Not enough storage
- ▶ Might miss some events
- ▶ Maintenance needed



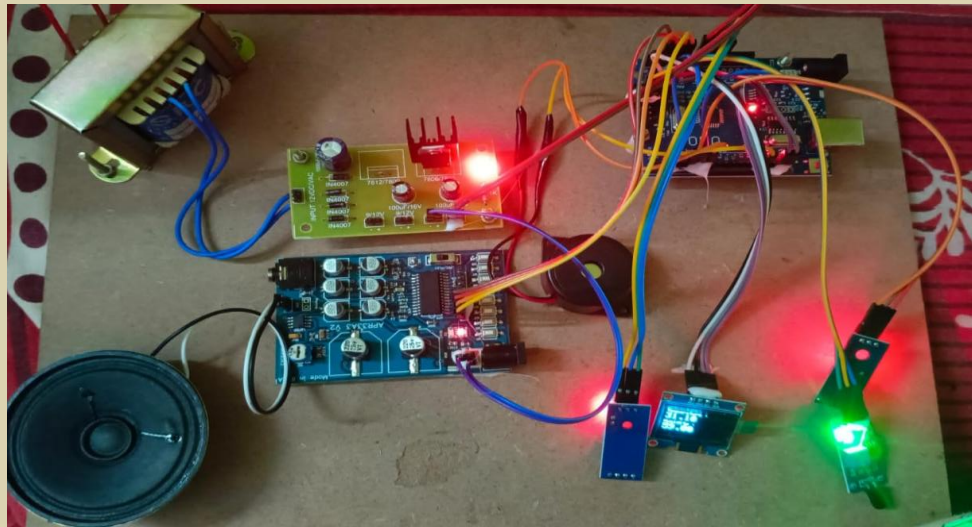
Applications

- ▶ Accident Investigation
- ▶ Watch how people drive
- ▶ Avoid accidents
- ▶ Find stolen cars
- ▶ Gives proof for insurance
- ▶ Emergency response



result

- ▶ Intelligent Event Data Recorders (IEDRs) are very helpful in vehicles. They give important data during accidents, help drivers improve, support insurance claims, and increase road safety. Their use makes driving safer, more secure, and smarter.



conclusion

- ▶ Intelligent Event Data Recorders (IEDRs) play an important role in modern vehicles. They help improve road safety, support accident investigations, and assist drivers and companies in many ways. Though there are some drawbacks, the overall benefits make IEDRs a valuable tool for safer and smarter driving.



reference

- 1. Research Paper "Design of Event Data Recorder (EDR) for Intelligent Transportation Systems." IEEE Conference Publication.
- 2. Book Navet, N., & Simonot-Lion, F. Automotive Embedded Systems Handbook. CRC Press.
- 3. Technical Paper Texas Instruments. "Automotive Black Box using Microcontroller."

